

The Exact Lipschitz Constant for an Absolute-Value Exponential Tilt of the Gaussian

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Abstract

Let γ_1 be standard Gaussian measure and, for $L > 0$, let ν_L have density proportional to $e^{L|t|}$ with respect to γ_1 . We calculate the global Lipschitz constant of the monotone transport T_L from γ_1 to ν_L . It is

$$\text{Lip}(T_L) = T'_L(0) = 2e^{L^2/2}\Phi(L),$$

and the continuous derivative attains this value only at the origin. The proof is an elementary comparison of Gaussian tails after an exact shift of the target quantile. We also give the resulting logarithmic asymptotic, including its first three Mills-ratio terms.

1 Statement and context

Write

$$\varphi(x) = \frac{1}{\sqrt{2\pi}}e^{-x^2/2}, \quad \Phi(x) = \int_{-\infty}^x \varphi(s) ds, \quad \bar{\Phi} = 1 - \Phi.$$

For $L > 0$, set

$$Z_L = \int_{\mathbb{R}} e^{L|t|} d\gamma_1(t), \quad q_L(t) = Z_L^{-1}e^{L|t|}\varphi(t),$$

and let ν_L be the probability measure with density q_L . If F_L is its distribution function, the canonical monotone transport is

$$T_L = F_L^{-1} \circ \Phi.$$

This is the one-dimensional Brenier map, taken in its everywhere-defined monotone representative.

This family occurs in Remark 5.7 of Gwóźdz [1]. That remark computes $T'_L(0) = Z_L$ and uses it as a lower bound for $\text{Lip}(T_L)$; the result below supplies the matching global upper bound and classifies equality. Our comparison is confined to this explicit one-dimensional family. A similar need to distinguish an exact constant on a specified model class from an unrestricted ambient constant is emphasized, in an unrelated spectral-set problem, by Zeng [2]. The latter reference supplies methodological context only and no optimal-transport claim. We make no priority claim beyond this comparison with the cited sources.

Theorem 1. *For every $L > 0$,*

$$\text{Lip}(T_L) = T'_L(0) = Z_L = 2e^{L^2/2}\Phi(L).$$

Moreover,

$$\{x \in \mathbb{R} : T'_L(x) = \|T'_L\|_{L^\infty(\mathbb{R})}\} = \{0\}.$$

As $L \rightarrow \infty$,

$$\log \text{Lip}(T_L) = \frac{L^2}{2} + \log 2 - \frac{\varphi(L)}{L} \left(1 - \frac{1}{L^2} + \frac{3}{L^4} + O(L^{-6}) \right). \quad (1)$$

Here the displayed maximizer set refers to the continuous derivative of the canonical representative. In particular, the isolated maximizer is retained even though it has Lebesgue measure zero.

2 Exact tail reduction and proof

Completing the square on the positive half-line gives

$$Z_L = 2 \int_0^\infty e^{Lt} \varphi(t) dt = 2e^{L^2/2} \Phi(L). \quad (2)$$

The source and target densities are continuous and strictly positive. Consequently, their distribution functions have continuously differentiable inverses, T_L is strictly increasing and C^1 , and

$$T'_L(x) = \frac{\varphi(x)}{q_L(T_L(x))}. \quad (3)$$

Both measures are symmetric, so uniqueness of the monotone rearrangement shows that T_L is odd. Thus $T_L(0) = 0$, $T_L(x) \geq 0$ for $x \geq 0$, and (3) yields

$$T'_L(0) = Z_L. \quad (4)$$

Fix $x \geq 0$ and put

$$y = T_L(x) - L.$$

For $t \geq 0$, another completion of the square and (2) give

$$q_L(t) = \frac{\varphi(t-L)}{2\Phi(L)}.$$

The monotone transport matches the upper-tail mass above x with that above $T_L(x)$. Hence

$$\overline{\Phi}(y) = 2\Phi(L)\overline{\Phi}(x). \quad (5)$$

Using the same density formula in (3), we also obtain

$$T'_L(x) = 2\Phi(L) \frac{\varphi(x)}{\varphi(y)}.$$

Together with (2), this is the exact identity

$$\frac{T'_L(x)}{Z_L} = \exp \left(\frac{y^2 - x^2 - L^2}{2} \right). \quad (6)$$

It remains to control the exponent. Since $T_L(x) \geq 0$, we have $y \geq -L$. Suppose first that $-L \leq y \leq 0$. Then

$$y^2 - x^2 \leq y^2 \leq L^2.$$

If $x > 0$, strict monotonicity and $T_L(0) = 0$ imply $T_L(x) > 0$, so $y > -L$ and the second inequality is strict. Suppose instead that $y \geq 0$. Because $L > 0$, we have $2\Phi(L) > 1$. Equation (5) then gives

$\bar{\Phi}(y) > \bar{\Phi}(x)$. The Gaussian upper tail is strictly decreasing, and therefore $0 \leq y < x$; in particular, $y^2 - x^2 < 0 < L^2$.

At $x = 0$, one has $y = -L$, so equality holds in (6). The preceding two cases prove

$$T'_L(x) \leq Z_L \quad (x \geq 0), \quad T'_L(x) = Z_L \iff x = 0.$$

The derivative of an odd differentiable function is even, so these conclusions hold on all of $\mathbb{R}$. The mean value theorem gives $\text{Lip}(T_L) \leq Z_L$, while difference quotients at the origin and (4) give the reverse inequality. This proves the exact Lipschitz constant and the asserted full maximizer set.

For completeness, we derive the stated asymptotic with a controlled remainder. For $k \geq 0$, integration by parts gives

$$\int_L^\infty \frac{\varphi(t)}{t^{2k}} dt = \frac{\varphi(L)}{L^{2k+1}} - (2k+1) \int_L^\infty \frac{\varphi(t)}{t^{2k+2}} dt.$$

Three applications, with the final positive integral retained, imply

$$\bar{\Phi}(L) = \frac{\varphi(L)}{L} \left(1 - \frac{1}{L^2} + \frac{3}{L^4} + O(L^{-6}) \right). \quad (7)$$

Indeed, the final integral is at most $L^{-6}\bar{\Phi}(L) \leq \varphi(L)L^{-7}$. If $r_L = \bar{\Phi}(L)$, then

$$\log \Phi(L) = \log(1 - r_L) = -r_L + O(r_L^2).$$

Moreover $r_L^2 = o(\varphi(L)L^{-7})$, so the quadratic term is absorbed by the remainder in (7). Substitution into the exact identity

$$\log Z_L = \frac{L^2}{2} + \log 2 + \log \Phi(L)$$

proves (1) and completes the proof of Theorem 1.

Remark 1. *The restriction $L > 0$ is used in the strict comparison $2\Phi(L) > 1$. At the excluded boundary value $L = 0$, the transport is the identity, $Z_0 = 1$, and every point maximizes the derivative.*

Assistance disclosure

The argument is analytic and uses no numerical experiment. Generative-AI assistance was used to help organize the exposition and to check the algebra, sign split, equality cases, and asymptotic remainder. The author reviewed the resulting mathematical statements and proof.

References

- [1] Maja Gwózdź. Caffarelli estimates under lipschitz perturbations, 2026. arXiv:2609.04052v1, 3 September 2026. <https://arxiv.org/abs/2609.04052>.
- [2] Zijian Zeng. Fixed convex-lens spectral constants: Möbius reduction, sharp model theorems, and angle-dependent bounds. Preprints.org, version 1, 2026. Posted 19 August 2026. doi:10.20944/preprints202608.1272.v1.